

Router Networking Lab

1) **Title:** Inter-Network communication using Router in Cisco Packet Tracer.

2) **Aim / Objectives :**

To establish communication between two different networks using a router.

3) **Devices Used :**

- 1 Router (2911)
- 2 Switches (2960)
- 2 PCs
- Copper Straight-through Cables

4) **IP Addressing Table :**

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.10	255.255.255.0
Router	G 0/0	192.168.1.1	255.255.255.0
Router	G 0/1	192.168.2.1	255.255.255.0
PC1	FastEthernet0	192.168.2.10	255.255.255.0

5) **Default Gateway Table :**

Device	Default Gateway
PC0	192.168.1.1
PC1	192.168.2.1

6) Procedure / Steps :

- Added one router, two switches, and two PCs in Packet Tracer.
- Connected all devices using straight-through cables.
- Configured router interfaces with IP addresses.
- Enabled interfaces using no shutdown.
- Assigned IP addresses and default gateway to PCs.
- Verified connectivity using ping command.

7) Router Configuration Commands :

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
```

8) Command Used :

```
ping 192.168.2.10
```

9) Result :

Successful communication established between two different networks using a router.

10) Concepts Learned :

- Routing basics
- Router functionality
- Default gateway
- Inter-Network Communication
- IP addressing

- Network Troubleshooting

11) Screenshot :

